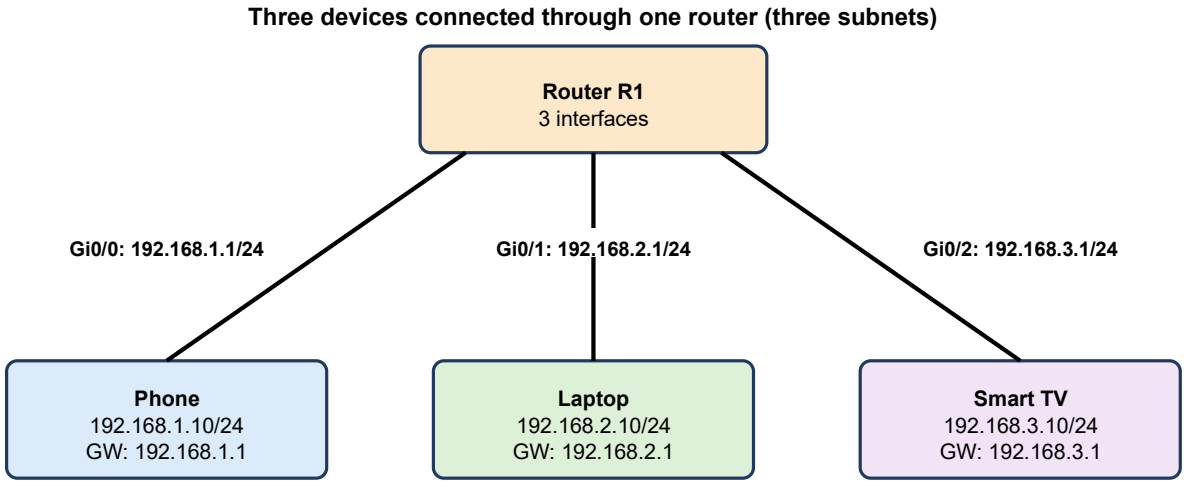
IP ROUTING AND ROUTING PROTOCAL

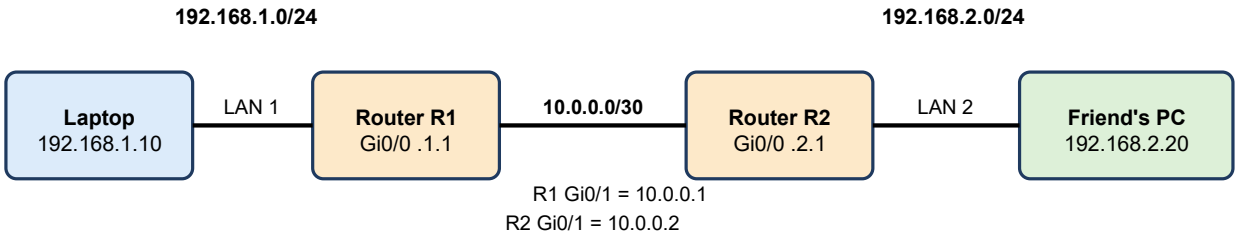
TASK-1 Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



Device / Interface	IP address	Subnet mask	Default gateway
Phone	192.168.1.10	255.255.255.0	192.168.1.1
Laptop	192.168.2.10	255.255.255.0	192.168.2.1
Smart TV	192.168.3.10	255.255.255.0	192.168.3.1
Router R1 Gi0/0	192.168.1.1	255.255.255.0	-
Router R1 Gi0/1	192.168.2.1	255.255.255.0	-
Router R1 Gi0/2	192.168.3.1	255.255.255.0	-

Each router interface is the default gateway of its own subnet. The router forwards packets between the three subnets.

TASK- 2 Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.



Packet: Laptop (192.168.1.10) sends data to Friend's PC (192.168.2.20), which is in a different subnet.

Routing tables used

Device	Destination network	Next hop	Interface
Laptop	0.0.0.0/0 (default gateway)	192.168.1.1	LAN
R1	192.168.1.0/24	Directly connected	Gi0/0
R1	10.0.0.0/30	Directly connected	Gi0/1
R1	192.168.2.0/24	10.0.0.2 (R2)	Gi0/1
R2	10.0.0.0/30	Directly connected	Gi0/1
R2	192.168.2.0/24	Directly connected	Gi0/0
R2	192.168.1.0/24	10.0.0.1 (R1)	Gi0/1

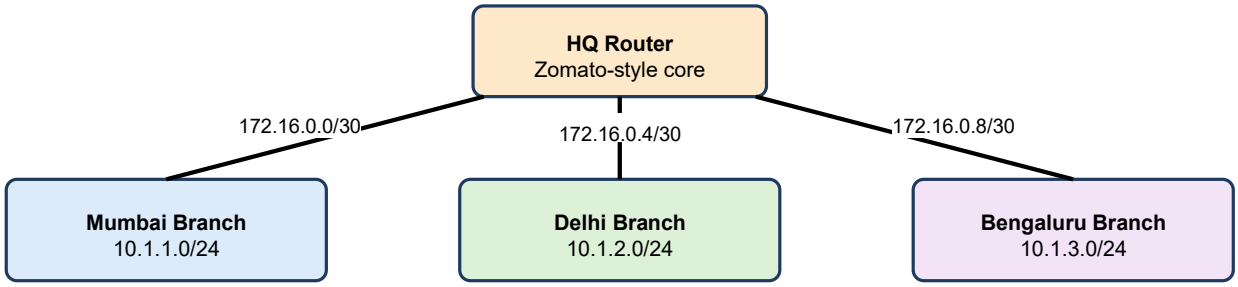
Step-by-step packet journey

1. **Laptop decision:** the laptop compares 192.168.2.20 with its own network using the mask: 192.168.2.20 AND 255.255.255.0 = 192.168.2.0, which is not 192.168.1.0. The destination is remote, so the packet goes to the default gateway 192.168.1.1 (R1). The laptop uses ARP to find R1's MAC address and sends the frame to it.
2. **Hop 1, Router R1:** R1 receives the packet on Gi0/0, reduces TTL by 1, and checks its routing table. The best match for 192.168.2.20 is 192.168.2.0/24 with next hop 10.0.0.2 via Gi0/1. R1 finds R2's MAC with ARP and forwards the packet.

3. **Hop 2, Router R2:** R2 receives the packet on Gi0/1, reduces TTL by 1, and checks its table. 192.168.2.0/24 is directly connected on Gi0/0, so R2 uses ARP to find the MAC of 192.168.2.20 and delivers the frame.
4. **Destination:** Friend's PC receives the packet. The reply travels the reverse way: PC to R2 (its gateway 192.168.2.1), then to R1 using the route to 192.168.1.0/24 via 10.0.0.1, then to the laptop.

Routing decision rule: every router picks the route with the longest prefix match (most specific network) for the destination address, then forwards the packet to that route's next hop or interface.

TASK-3 Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.



Each branch has its own LAN subnet and a point-to-point link to the HQ router.

Routing table of the HQ router

Destination network	Subnet mask	Next hop	Interface	Type
10.1.1.0 (Mumbai LAN)	255.255.255.0	172.16.0.2	Gi0/0	Static
10.1.2.0 (Delhi LAN)	255.255.255.0	172.16.0.6	Gi0/1	Static
10.1.3.0 (Bengaluru LAN)	255.255.255.0	172.16.0.10	Gi0/2	Static
172.16.0.0 (HQ-Mumbai link)	255.255.255.252	Directly connected	Gi0/0	Connected
172.16.0.4 (HQ-Delhi link)	255.255.255.252	Directly connected	Gi0/1	Connected
172.16.0.8 (HQ-Bengaluru link)	255.255.255.252	Directly connected	Gi0/2	Connected

Example: the router in the Mumbai branch

Destination network	Subnet mask	Next hop	Interface
10.1.1.0	255.255.255.0	Directly connected	Gi0/1 (LAN)
172.16.0.0	255.255.255.252	Directly connected	Gi0/0 (to HQ)
0.0.0.0 (default)	0.0.0.0	172.16.0.1	Gi0/0

The default route sends all traffic for Delhi, Bengaluru and the Internet to HQ, which then uses its table above.

TASK-4 Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

	Static routing	Dynamic routing (RIP / OSPF)
Pros	1. No routing protocol traffic, and almost no CPU ormemory use. 2. Predictable and more secure, because the admincontrols every path. 3. Very simple for small networks, e.g. a homerouter with one default route to the ISP.	1. Routes update automatically when a link fails or isadded, e.g. a food delivery app's network can reroute around a failed branch link. 2. Scales well to many routers and branches. 3. OSPF converges quickly and can pick the bestpath by cost.
Cons	1. Does not adapt to failures, so traffic is lost until anadmin fixes the route. 2. Hard to manage in large networks, because everyroute is typed by hand. 3. Human mistakes (wrong next hop) are commonas the network grows.	1. Uses router CPU, memory and bandwidth forprotocol updates. 2. More complex to configure and troubleshoot. 3. Without authentication, fake routing updates canbe accepted, which is a security risk.

Real-world examples

- **Static:** the Wi-Fi router at home only needs one default route to the ISP, which is how phone apps such as Zomato or Swiggy reach the Internet.
- **Dynamic:** a delivery company with many city offices and data centres uses OSPF so that if one link goes down, orders and rider tracking traffic automatically move to another path.
- **RIP vs OSPF:** RIP is simple but slow and limited to 15 hops. OSPF is faster, scalable and the usual choice for company networks.

TASK-5 Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does. *Hint: Paste the AI-generated config and annotate each part with a short description.*

AI tool used: Claude (Anthropic). Prompt: "Give a sample OSPF configuration for two routers connected by a link, with one LAN behind each router." Topology: R1 (LAN 192.168.1.0/24) is connected to R2 (LAN 192.168.2.0/24) through the link 10.0.0.0/30.

Router R1 configuration with annotations

Configuration line	What it does
enable	Enters privileged EXEC mode on the router.
configure terminal	Enters global configuration mode so settings can be changed.
hostname R1	Sets the router name to R1.
interface GigabitEthernet0/0	Selects the LAN-facing interface.
ip address 192.168.1.1 255.255.255.0	Gives the interface the LAN gateway address.
no shutdown	Turns the interface on (interfaces are off by default).
interface GigabitEthernet0/1	Selects the link that goes to R2.
ip address 10.0.0.1 255.255.255.252	Sets the router-to-router link address (/30 has 2 usable hosts).
router ospf 1	Starts OSPF process number 1 (the number only matters locally).
router-id 1.1.1.1	Sets a unique ID for R1 in the OSPF network.

network 192.168.1.0 0.0.0.255 area 0	Enables OSPF on interfaces in the LAN network and advertises it. 0.0.0.255 is a wildcard mask; area 0 is the backbone area.
network 10.0.0.0 0.0.0.3 area 0	Enables OSPF on the link to R2 so the two routers can become neighbours.
passive-interface GigabitEthernet0/0	Stops OSPF hellos on the LAN (no routers there), but the LAN is still advertised.
end / write memory	Returns to privileged mode and saves the configuration.

```
! ---------- R1 ---------enable
configure terminal hostname R1
interface GigabitEthernet0/0  ip
address 192.168.1.1 255.255.255.0
no shutdown interface
GigabitEthernet0/1  ip address
10.0.0.1 255.255.255.252  no
shutdown router ospf 1  router-id
1.1.1.1

 network 192.168.1.0 0.0.0.255 area
0  network 10.0.0.0 0.0.0.3 area 0
passive-interface GigabitEthernet0/0
end write memory

! ---------- R2 ---------enable
configure terminal hostname R2
interface GigabitEthernet0/0  ip
address 192.168.2.1 255.255.255.0
no shutdown interface
GigabitEthernet0/1  ip address
10.0.0.2 255.255.255.252  no
shutdown router ospf 1  router-id
2.2.2.2

 network 192.168.2.0 0.0.0.255 area
0  network 10.0.0.0 0.0.0.3 area 0
passive-interface GigabitEthernet0/0
end write memory
```